

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 10

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • Harvest rate below the maximum sustainable yield; only economically valuable species (mahogany) selectively logged, not clear-felled (1) • Trees felled only where other mahoganies stand upwind so seeds recolonise the clearing; large seed trees and small trees left to grow (1) • Buffer zones with no harvesting or machinery near rivers / wetlands (1) • Restricted machinery and planned track routes re-used to avoid damaging roots of retained / rare / fruit trees (1) • Each felled tree tagged and tracked to the retailer for FSC accreditation; rangers control illegal logging; tree nurseries replant over-exploited and ecologically important species (1)	5
01.2	<i>1 mark for each point.</i> • The reserve removed traditional uses (fuelwood, timber, farming clearings, hunting, medicines) so people would otherwise resent / ignore it (1) • Employment as rangers, forestry and tourism workers; ecotourists use local restaurants, produce and souvenirs; offcuts given to local carvers (1) • Benefits give the community an incentive to protect the forest; carbon sequestration payments from companies fund management (1)	3
01.3	<i>1 mark for each point.</i> • Removing individual trees leaves small clearings so remaining populations survive nearby and the forest structure / microclimate is kept; timber resource used over a longer period (1) • Disadvantage: more labour-intensive and expensive; more tracks needed per tree (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 369, 386-387; AQA spec 3.5.3.3.	

Question	Marking guidance	Marks
02.1	<i>1 mark for each point.</i> - Domesticated crops were bred from a small number of original plants so they have a small gene pool / low genetic diversity (1) - Selective breeding for yield, appearance or shelf life may have lost other useful genes (eg taste, disease resistance) (1) - Genetically similar plants share the same susceptibilities so a new disease or environmental change can affect the whole crop (1)	3
02.2	<i>1 mark for each point.</i> - A region with a high concentration of the wild relatives of important crops - usually where the crop was first domesticated (eg wheat / barley in the Middle East; potato, tomato in South America; rice in India; maize in Central America) (1) - They hold the largest gene pools / most genetic variety for future crop breeding (disease resistance, drought and salt tolerance, yield) (1) - Many are in regions where environmental degradation / habitat loss threatens the wild plants (1)	3
02.3	<i>1 mark for each point.</i> - A large population may all descend from a few closely related individuals so they are genetically similar (1) - Similar adaptations mean all are susceptible to the same change (disease, climate) (1) - Greater risk of inbreeding so disadvantageous recessive genes are expressed (eg amelanism, poor muscle development in lion cubs); so species should be protected across their whole range to retain regional genes (1)	3
02.4	<i>1 mark.</i> Moroma bean (drought resistant, soya-like protein); yeheb tree (drought resistant nuts, poor soils); potato bean (high protein); perennial maize / Kernza (no annual replanting, less ploughing / erosion); salt-tolerant spinach relatives (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 15-17; AQA spec 3.1.2.1.2.	
03.1	<i>2 marks for each point (partial credit for a partly correct point).</i> - Size: shows whether the population is viable / how urgent action is; sets a baseline to judge success; needed to estimate carrying capacity and sustainable off-take (2) - Distribution: shows which habitats / areas are used and whether the population is fragmented into non-viable isolated groups needing corridors; identifies key areas to protect (2) - Survival rate (mortality): identifies the life stage and causes of death (predation, disease, poaching) so management targets the limiting factor (2) - Age structure: many young indicates a growing population; few young / ageing population indicates breeding failure or high juvenile mortality even if adults are numerous; k-selected species need breeding adults protected (2)	8
03.2	<i>1 mark for each point.</i> - Direct counts from vehicles / aircraft / drones along transects, or camera traps with individual recognition (1) - Sub-sample counts in known areas scaled up to the whole reserve; mark-release-recapture (Lincoln index) for smaller species (1)	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 400-402; AQA spec 3.1.2.3.5.	
04.1	<i>1 mark for each point.</i> • Chicken and salmon need far less feed per kg gained (1.2-1.8 vs 7.0) (1) • Fish are ectotherms so no energy is used for thermoregulation and water supports their mass; poultry grow fast and are kept warm indoors with movement restricted (1) • Cattle are large endotherms, use energy in movement, and lose energy in methane from cellulose digestion (1)	3
04.2	<i>1 mark for each point.</i> • Control of movement (housing / restricted grazing) so less energy is used in muscle activity (1) • Temperature control (heated / insulated buildings) so less energy is used in maintaining body temperature (1) • Species selection: choose species with low FCRs (poultry, fish); high-energy easily digested diets; slaughter at the fastest growth stage (1)	3
04.3	<i>1 mark for each point.</i> • Only about 10% of the energy at one trophic level is passed to the next; most is lost as heat in respiration / metabolism or in undigested material (1) • Humans eating plants (primary consumers) receive more of the energy captured by photosynthesis than eating animals fed on those plants (secondary consumers) (1) • Livestock better where land cannot grow crops (upland, semi-arid, steep, thin soils): grazing animals with symbiotic gut bacteria digest cellulose in grass that humans cannot (1) • Omnivores (pigs) convert food wastes (dairy by-products, catering waste) into meat (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 326-328; AQA spec 3.5.1.2.	
05.1	<i>Correct answer only.</i> • Its mass (1)	1
05.2	<i>1 mark for each point.</i> • The 24-hour rotation period means the surface is not exposed to sunlight for long, so temperature extremes / excessive heating and cooling are reduced (1) • The axis is tilted relative to the orbit so the amount of insolation at each latitude changes through the year, producing seasonal variations in conditions (1)	2
05.3	<i>1 mark for each point.</i> • The distance from the Sun gives surface temperatures between 0 and 35 °C in most areas: too close and water would boil / evaporate, too far and it would freeze (1) • Atmospheric pressure (produced by the mass of gases held by gravity) is high enough to stop the water boiling / evaporating away; at much lower pressure there would be no liquid water (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-7; AQA spec 3.1.1.1.	

Question	Marking guidance	Marks
06.1	<i>1 mark for each point.</i> - Negative feedback responds to any change by re-establishing the previous equilibrium (1) - Only large unusual events overwhelm the feedback so a new equilibrium forms (1) - Example: methane hydrate release at the end of the Permian causing climate change and mass extinction (1)	3
06.2	<i>1 mark for each point.</i> - In a diverse system each component is relatively less important so the loss / change of one has a small effect; alternative food sources and relationships buffer the change (1) - Natural: coral reefs and tropical rainforests with complex food webs are stable; simple polar / desert systems fluctuate (1) - Human: agroecosystems focus on a few crops and remove detritivores, pollinators and predators so are vulnerable to pests, disease, weather (1) - Diverse technical systems: a society using a range of renewable energy resources is more secure than one relying on a single source (1)	4
06.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> - Melting permafrost releasing methane (1) - Reduced albedo as ice / snow melt (1) - Methane hydrate release (1) - Forest and peat fires (1) - Faster soil decomposition (1) - Ocean acidification reducing sequestration (1) - Cirrus cloud formation (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 371-372; AQA spec 3.6.1.	
07.1	<i>1 mark for each point.</i> - MEDCs: market demand and higher prices for the flavour of carnivorous fish (salmon, trout) justify high inputs (fishmeal, cooler climates suit them) (1) - LEDCs: herbivores / omnivores feed on natural vegetation, phytoplankton and wastes so need low inputs; warm climates suit tilapia (1) - Herbivores are more energy-efficient (lower trophic level) so more food per unit area (1)	3
07.2	<i>1 mark for each point.</i> - Some genetically female (XX) fish are treated with testosterone so they become functional males producing only X-carrying sperm (1) - Crossed with normal females (XX) all offspring are XX females (1) - Female trout meat has a better flavour; in tilapia the reverse (all males) because males grow larger / more efficiently (1)	3
07.3	<i>1 mark for each point.</i> - Salmon need high oxygen: water sprays / weirs aerate; wastes removed to prevent deoxygenation (1) - Constant circular flow keeps fish swimming the same way - fewer collisions / injuries, more muscular fish, higher stocking density (1) - Artificial lighting delays maturation (salmon stop growing at maturity) so larger fish; controls smoltification (1) - Cool sites (Scotland, Norway) because warm water holds less oxygen although warmth increases growth (1)	4

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 352-355; AQA spec 3.5.2.3.	
08.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • Drainage makes soils aerobic and drier, increasing grass growth but wetland plants and the animals they support (waders, amphibians, invertebrates) cannot survive (2) • Fertiliser increases nutrient availability so vigorous species (grasses, nettles) grow taller and out-compete smaller wildflowers for light; diversity falls (2) • Hedgerow removal destroys habitat / corridors and nesting sites; loss of natural pest predator habitats increases pest damage; larger fields increase wind erosion; genetic contamination and introduced species also accepted (2)	6
08.2	<i>1 mark for each point.</i> • Long-continued traditional farming has created plagioclimax habitats colonised by many species: hay meadows, chalk grassland, grazed heath / moorland, hedgerows (1) • These communities depend on the farming continuing (1)	2
08.3	<i>1 mark for each point.</i> • CO₂ from fossil fuel use in machinery and embodied energy of fertiliser (Haber process) and machinery; ploughing increases soil respiration (1) • Methane from livestock intestines and rice paddies; nitrous oxide from manure and nitrogen fertilisers (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-330; AQA spec 3.5.1.3.	
09.1	<i>1 mark for each point.</i> • Tropical intertidal coastal forests of halophytic (salt-tolerant) trees in saline, oxygen-deficient (anaerobic) mud (1) • Prop / aerial roots for support in soft sediment and for gas exchange (pneumatophores); salt excretion (1) • Seeds germinate on the parent tree and float, rooting where they touch sand - rapid natural regeneration (1)	3
09.2	<i>1 mark for each point.</i> • Coastal erosion / storm protection: roots and vegetation absorb wave energy (least tsunami damage in 2004 where mangroves were intact) (1) • Fisheries: nursery grounds where dense roots protect young fish that later live on reefs / open sea (1) • Protection of coral reefs: slow river water so suspended solids settle and nutrients are absorbed before reaching reefs (1) • Resources: timber, fuel; medicines (antimicrobials against MRSA, cyanobacterium for lung cancer); high biodiversity (crabs, shrimps, sponges, birds) (1)	4
09.3	<i>1 mark for each point.</i> • Clearance for aquaculture (shrimp ponds), urban development, ports, tourism; pollution; sea level rise / climate change; destruction of coral reefs that shelter the coast (1) • Legal protection / designated areas (eg Sri Lanka protected all remaining mangroves in 2015) (1) • Replanting and natural regeneration; control of damaging activities; Ramsar sites (1)	3

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 66-67; AQA spec 3.1.2.3.4.	
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Water: catchment management, conservation, metering, greywater, re-use of treated effluent, aquifer recharge, sustainable urban drainage - largely cyclical already; limits in arid regions • Minerals: exploiting low-grade ores with low-temperature processes (bioleaching), extending product lifespans, repair / re-use / repurpose / recycle, design for disassembly, reduced packaging; limits: dispersal, alloys, energy of collection • Waste management: repair, re-use, refurbish, recycle, composting and anaerobic digestion returning nutrients, minimal landfill, energy from residual waste; EU targets on packaging, batteries, WEEE • Pollution control: shift from post-production treatment to non-release - fuel desulfurisation not FGD, fuel cells / electric vehicles not combustion, industrial symbiosis (Kalundborg: waste heat, gypsum, fly ash, yeast used by neighbours); limits for legacy pollution • Energy: renewables, storage, energy density technologies, low-temperature manufacturing, CHP; limits of intermittency, transport fuels • Food: natural processes for nutrients (legumes, organic matter), pest control (IPM), soil biota, agri-environment schemes, IMTA aquaculture; tension with maximum yield for a growing population • Land uses supporting ecosystems: urban green infrastructure, gardens, permeable surfaces, habitat in redeveloped docks / lagoons • Judgement: high applicability in some sectors (water, waste), partial in others (minerals, energy for transport); needs policy support	9
10.2	1 mark. • Locating industries so that the wastes of one (heat, gases, by-products) become the raw materials of another, mimicking natural cycles (eg Kalundborg eco-industrial park) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 375-379; AQA spec 3.6.4.1.	

Question	Marking guidance	Marks
11.1	Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Importance: resources (timber, fuel, food, fibres, medicines) and ecosystem services (atmospheric regulation, carbon store, hydrological regulation, climate, soil conservation, habitat, recreation) • Causes: agriculture (subsistence and cash crops), mineral extraction, reservoirs, urbanisation, transport, unsustainable timber, fuel and fodder exploitation above MSY; trends by region • Consequences: loss of resources, biodiversity and species, fragmentation; hydrological changes; soil impacts; climate impacts (albedo, carbon, rainfall) • Productivity vs biodiversity: non-indigenous species, monocultures, close planting, simple age structure, clear-felling • Sustainable management: MSY harvest rates, mixed indigenous species, mixed age structure, selective logging, traditional coppicing / pollarding, FSC certification, community involvement, protected areas, debt-for-nature, afforestation programmes (China); Rio Bravo case study • Evaluation: economics of slow-growing hardwoods, enforcement, demand from MEDCs, ITTO failures; judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-57, 360-369, 386-387; AQA spec 3.5.3, 3.1.2.3.4.	25

Question	Marking guidance	Marks
11.2	Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Temperate broadleaf woodland: seasonality, deep soils, high biodiversity, resources, climate and soil regulation; deforestation, fragmentation, management change; SSSIs, ancient woodland protection, coppicing, mixed planting • Tropical rainforest: stable warm wet climate, stratification, very high biodiversity, carbon and hydrological roles, thin soils; fuelwood, timber, agriculture, mining, reservoirs, climate change; protected areas, debt-for-nature swaps, sustainable exploitation • Tropical coral reefs: cnidarian-zooxanthellae symbiosis, light, temperature, turbidity, salinity; fisheries, erosion protection, medicines, tourism; damage, souvenirs, sedimentation, pollution, bleaching, fishing, introduced species; MPAs, no-take zones, control of activities • Deep-water reefs: cold, dark, slow growth; research, fisheries; trawling, oil / gas, acidification; SACs • Oceanic islands: isolation, endemism, no mammal predators; exploitation, introduced species, habitat change, sea level; eradication, control of development and visitors • Mangroves: halophytes, anaerobic mud; erosion protection, fisheries nurseries, sediment trapping; clearance for aquaculture / urban / ports, pollution, climate change; protection, replanting • Antarctica: extreme cold, seasonality, marine nutrients, ice; water store, albedo, carbon, research; climate change, ozone, tourism, fishing, minerals; Antarctic Treaty, CCAMLR, tourism and waste rules • Synthesis: features -> importance -> threats -> strategy; evaluation of success; conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-71; AQA spec 3.1.2.3.4.	25